THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS (PART 1 + PART 2)

Chapters 1–14 | Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 02	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The mirror image of the point (5, −2) in the x-axis is: **1**

(a) (−5, −2) (b) (5, 2) (c) (2, −5) (d) (−5, 2)

2. The midpoint of the segment joining A(−4, 7) and B(6, −3) is: **1**

(a) (−5, 5) (b) (1, 2) (c) (1, 5) (d) (5, 2)

3. If (k, 2) is a solution of the equation 3x + 4y = 20, then the value of k is: **1**

(a) 5 (b) 8 (c) 4 (d) 2

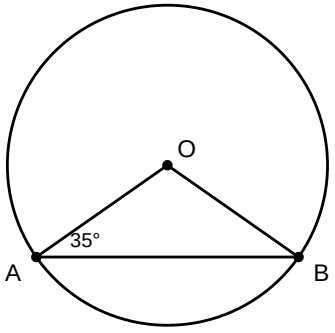
4. Which of the following has a terminating decimal expansion? **1**

(a) $\frac{5}{6}$ (b) $\frac{4}{15}$ (c) $\frac{9}{14}$ (d) $\frac{7}{40}$

5. The value of $51^2 - 49^2$ is: **1**

(a) 200 (b) 2 (c) 2500 (d) 100

6. In the figure, O is the centre of the circle and ∠OAB = 35°. Then ∠AOB is equal to: **1**



(a) 35° (b) 70° (c) 145° (d) 110°

7. A chord of length 16 cm is drawn in a circle of diameter 20 cm. Its distance from the centre is: **1**

(a) 12 cm (b) 8 cm (c) 2 cm (d) 6 cm

8. The area of an equilateral triangle of side 10 cm, obtained from Heron's formula, is: **1**

(a) $25\sqrt{3}$ cm² (b) $100\sqrt{3}$ cm² (c) $50\sqrt{3}$ cm² (d) 25 cm²

9. A bag contains 5 red, 3 green and 2 blue balls. A ball is drawn at random. The probability that it is not red is: **1**

(a) $\frac{1}{5}$ (b) $\frac{1}{2}$ (c) $\frac{2}{5}$ (d) $\frac{3}{10}$

10. The n-th term of a sequence is $t_n = n(n - 2)$. Then t_7 is: **1**

(a) 14 (b) 35 (c) 21 (d) 49

11. In which of the following are BOTH the proposition and its converse true? **1**

(a) If $x = y$, then $x^2 = y^2$.

(b) If a number is a multiple of 8, then it is a multiple of 4.

(c) If two triangles have the same area, then they are congruent.

(d) If n is a perfect square, then n has an odd number of factors.

12. A counterexample to the claim '$n^2 + n + 11$ is prime for every positive integer n' is: **1**

(a) n = 10 (b) n = 1 (c) n = 3 (d) n = 2

13. Jaspreet cycled for 5 days in each of three weeks. Her weekly averages were 12, 18 and 15 minutes. Her average for the 15 days is: **1**

(a) 15 minutes (b) 45 minutes (c) 14 minutes (d) 16 minutes

14. A cylindrical drum has radius 7 cm and height 10 cm. Its volume ($\pi = 22/7$) is: **1**

(a) 3080 cm³ (b) 770 cm³ (c) 1540 cm³ (d) 440 cm³

15. In the digit-by-digit addition algorithm the carry can never exceed 1 because the largest possible column total is: **1**

(a) 9 + 9 = 18 only (b) 10 + 10 = 20

(c) 9 + 9 + 9 = 27 (d) 9 + 9 + 1 = 19

16. The diagonals AC and BD of a parallelogram ABCD meet at O. If AC = 14 cm and BD = 10 cm, then OA + OB is: **1**

(a) 10 cm (b) 12 cm (c) 7 cm (d) 24 cm

17. The sides of a triangle are 10 cm, 14 cm and 18 cm. The perimeter of the triangle formed by joining the midpoints of its sides is: **1**

(a) 21 cm (b) 14 cm (c) 28 cm (d) 42 cm

18. The 10th term of the arithmetic progression 4, 9, 14, … is: **1**

(a) 44 (b) 54 (c) 49 (d) 45

19. **Assertion (A):** 0.333… (3 repeating) is a rational number. **1**
Reason (R): Every repeating decimal can be written in the form p/q, where p and q are integers and $q \neq 0$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** If a proposition is true, then its converse is always true. **1**
Reason (R): The proposition 'If it rains, then the road is wet' is true, but the road can be wet without rain.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

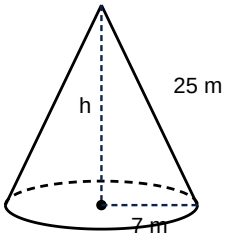
21. An e-rickshaw charges ₹30 for the first 2 km and ₹9 for every additional km. Find the fare for a journey of 8 km. **2**

22. Without actual division, decide whether $\frac{7}{125}$ has a terminating decimal expansion. If so, state the number of decimal places. **2**

OR

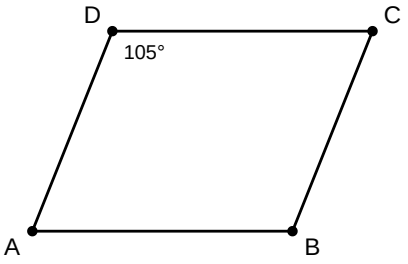
Find two rational numbers between $\frac{3}{8}$ and $\frac{1}{2}$.

23. A conical tent has base radius 7 m and slant height 25 m. Find its curved surface area and its height (π = 22/7). **2**



24. A dairy mixes 400 litres of milk containing 3% fat with 100 litres of milk containing 8% fat. Find the percentage of fat in the mixture. **2**

25. In the parallelogram ABCD shown, ∠ADC = 105°. Find ∠DAB and ∠ABC, giving reasons. **2**



SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express 2.3636… (36 repeating) in the form p/q in lowest terms. **3**

27. A survey of 200 families recorded the number of children in each family: **3**

Children	0	1	2	3
Families	30	80	70	20

A family is chosen at random. Find the probability that it has (i) no child, (ii) exactly one child, (iii) at least two children.

OR

A bag contains 7 red, 5 white and 8 black balls. One ball is drawn at random. Find the probability that it is (i) red, (ii) not white, (iii) white or black.

28. Draw the graphs of the equations 2x + y = 8 and x − y = 1 on the same grid and use them to find the solution of the pair. **3**

29. Write the converse of the proposition ‘If a quadrilateral is a rhombus, then its diagonals are perpendicular.’ Decide whether the converse is true and give a counterexample if it is not. **3**

30. A chord of length 12 cm is drawn in a circle of radius 10 cm. Find (i) its distance from the centre; (ii) the length of another chord of the same circle at a distance of 6 cm from the centre. **3**

OR

Using Heron's formula, find the area of a triangle with sides 15 cm, 36 cm and 39 cm.

31. Write an algorithm to find the lcm of two numbers m and n by testing the numbers max(m, n), max(m, n) + 1, … in turn. Execute it for m = 4 and n = 6. **3**

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. The three corners of a triangular flag are marked on a grid as A(−3, −2), B(5, −2) and C(1, 4). **5**
(a) Plot the points and join them. (2 marks)
(b) Find the lengths AB, BC and CA. (1½ marks)
(c) Name the type of triangle on the basis of its sides and find its area. (1½ marks)

33. How many multiples of 8 lie between 200 and 500? Find their sum. (2½ + 2½) **5**

OR

The 5th term of an AP is 17 and its 12th term is 38. Find the AP and the sum of its first 20 terms. (2 + 3)

34. Prove that if the diagonals of a quadrilateral bisect each other, then it is a parallelogram. **5**

OR

State and prove the converse of the Baudhāyana–Pythagoras theorem: if $a^2 + b^2 = c^2$ for the sides of a triangle, then it is right-angled.

35. The area of a rectangular carpet is $(8x^2 + 14x + 3)$ m² and its breadth is $(2x + 3)$ m. Factorise the expression to find its length. If $x = 3$, find the perimeter of the carpet. **5**

OR

The sum of the length and breadth of a rectangular notice-board is 17 dm and its area is 60 dm². Form a quadratic expression in the breadth x, factorise it and find the length and breadth.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A drone delivery company maps its city on a coordinate grid (1 unit = 100 m). The warehouse W is at the origin, the shop S is at (6, 8), the school T is at (−6, 8) and a house H is at (−3, −4).

(i) Find the distance of the shop S from the warehouse W (in grid units). 1

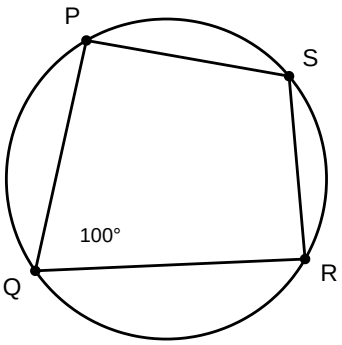
(ii) Find the coordinates of the midpoint of the segment ST. 1

(iii) Show that the warehouse is at the same distance from S and T. 2

OR

(iii) A drone flies directly from H to S. Find the distance it covers in metres.

37. Four floodlights P, Q, R and S are fixed on the boundary of a circular ground (in that order). It is known that ∠PQR = 100° and ∠QPS = 85°.



(i) Find ∠PSR. 1

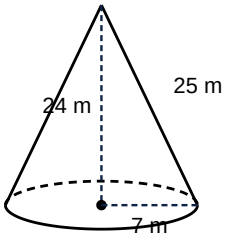
(ii) Find ∠QRS. 1

(iii) If ∠QPR = 40°, find ∠QSR and state the reason. 2

OR

(iii) The side PS is produced to a point T. Find the exterior angle ∠RST and state the property used.

38. A grain merchant stores grain in a conical heap of base radius 7 m and height 24 m (π = 22/7).



(i) Find the slant height of the heap. 1

(ii) Find the volume of grain in the heap. 1

(iii) Find the area of canvas needed to cover the heap. 2

OR

(iii) The grain is poured into a cylindrical bin of the same base radius. Find the height of grain in the bin.